

Inside Water

When the Slower Player Can Cut Off the Faster

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Two players race toward a fixed goal. Each has a known constant speed and a freely chosen direction. A player wins by being collinear and in front: reaching the goal first or getting between the opponent and the goal (“inside water”). We prove that the slower player sometimes has an inside water strategy that overturns the direct race — while the faster player never does. We classify all configurations into four regimes: slower wins direct, slower wins by guaranteed inside water, contested via sidestep (mixed Nash equilibrium), and faster wins. The faster player’s only recourse is the sidestep — deviating from the direct path to avoid being cut off — but in the guaranteed regime, sidestepping cannot change the outcome. Multi-step and limit strategies collapse to the Nash equilibrium of the one-shot game.

Motivation

In water polo, a one-on-one between an attacker and a defender often comes down to who establishes *inside water* relative to the goal. If the attacker gets in front of the defender with a clear line to the goal, the outcome is effectively decided: the attacker faces only the goalkeeper, and any physical interference from behind results in an exclusion or penalty — both advantageous for the attack. Conversely, if the defender cuts off the attacker, the attack is neutralised. Unlike land sports, where a player can plant a foot and make a sharp cut, water cannot be pushed against easily. Direction changes in water are slow. Once a player is in front, the trailing player cannot realistically overtake.

This makes inside water a *terminal condition*: the first player to establish inside water wins the play. The game reduces to a race — not just to the goal, but to a strategic point between the opponent and the goal.

The same structure appears beyond sports. In military interception, a slower unit may attempt to position itself between a faster unit and its objective. In autonomous vehicle coordination, one agent may need to merge into another’s lane before a bottleneck. In competitive robotics, two agents may race to a resource point. In each case, the question is the same: can a player that loses the direct race win by positioning?

Setup

Fix a plane with a goal X , a faster player B (speed v_B), and a slower player A (speed $v_A \leq v_B$),

without loss of generality. Each chooses a direction and moves in a straight line at full speed. The speed ratio is $r = v_A/v_B$, $r \in (0, 1]$.

A player wins by being collinear and in front: at some time t , $A(t)$ lies on the open segment from $B(t)$ to X . All information is public: positions, speeds, and directions are known to both.

Place X at the origin, B at distance d_B from X . Define the unit direction from B toward X :

$$\hat{u} = (X - B)/d_B, \quad d_B = \|X - B\| \quad (1)$$

The slower player A is at distance $d_A = \|X - A\|$ from X . Decompose $A - B$ into a component along $\hat{u}$ and a perpendicular component:

$$p = (A - B) \cdot \hat{u}, \quad \mathbf{n} = (A - B) - p\hat{u}, \quad h = \|\mathbf{n}\| \quad (2)$$

so p is the signed projection of $A - B$ toward X , and h is the perpendicular distance from A to the line through B and X .

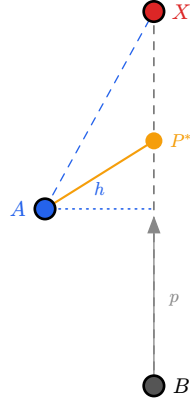


Figure 1: Setup. A (slower) may go direct to X or to the cut-off point P^* on B 's direct path. h : perpendicular distance from A to line BX . p : projection of $A - B$ toward X .

The Direct Race

If both swim straight to X , the arrival times are $T_A = d_A/v_A$ and $T_B = d_B/v_B$. A wins iff:

$$d_A < r d_B \quad (3)$$

A must be closer to X by a factor of at least r to compensate for the speed disadvantage. The locus of equal arrival times is the Apollonius circle, known since antiquity. Our contribution is not the race itself, but what happens when the race is lost.

Inside Water

Rather than racing to X , A targets a point on B 's direct path and arrives there before B — getting between B and X . The inside water strategy assumes the opponent swims direct, and depends only on positions and speeds (not on B 's actual direction).

The optimal interception satisfies $\cos \psi = r$, where ψ is the angle between $\overline{AP^*}$ and $\overline{BX}$ at P^* (Appendix A.2 B). This is the *Apollonius interception angle*: in any configuration where inside water is feasible, A approaches B 's path at the same angle, depending only on the speed ratio. The inside water point is:

$$P^* = B + (p + sh)\hat{u}, \quad s = r/\sqrt{1 - r^2} \quad (4)$$

The *inside water condition* is:

$$h\sqrt{1 - r^2} < rp \quad \text{and} \quad 0 < \lambda^* \leq 1 \quad (5)$$

where $\lambda^* = (p + sh)/d_B$ is the position of P^* along $\overline{BX}$. A 's inside water time is $\tau_A = h/(v_A\sqrt{1 - r^2})$.

The Inside Water Theorem

Theorem 1 (Inside water asymmetry). *The slower player (A) sometimes has a viable inside water strategy. The faster player (B) never does.*

(a) *If B loses the direct race ($T_A < T_B$), then B cannot cut off A : at every point on A 's direct path, A arrives first.*

(b) *Only the slower player can win by cutting off; the faster player's response (sidestep) is at best defensive.*

Proof sketch (full proof in Appendix A.1 A). Define $g(\mu) = \tau_B(\mu) - \tau_A(\mu)$, the arrival time difference at a point $Q(\mu)$ on A 's path ($\mu = 0$ at A , $\mu = 1$ at X). The derivative is:

$$g'(\mu) = \frac{d_A}{v_B} \left[\cos \psi(\mu) - \frac{1}{r} \right] \quad (6)$$

Since $\cos \psi(\mu) \leq 1 < 1/r$, the bracket is *always negative*. g is strictly decreasing, with minimum $g(1) = T_B - T_A$. When $T_A < T_B$, this minimum is positive: B arrives late everywhere on A 's path. $\square$

For A cutting off B , the equation is $\cos \psi = r < 1$, which *does* have a solution.

Why is this counterintuitive? One expects the faster player could use that speed to intercept the opponent's path. It fails because B 's interception geometry would require $\cos \psi = 1/r > 1$ — an angle that does not exist. The faster B is, the worse the geometry: speed carries B past any interception point.

Analogy: a jet trying to cut off a bicycle on a narrow path. The jet cannot decelerate into position — its speed overshoots. The bicycle, being slow, steps onto the runway at just the right angle.

The Sidestep

When A can cut off B 's direct path, B may deviate to avoid being cut off: the *sidestep*.

A sidestep of angle θ from the direct direction for duration δ increases B 's arrival time at X by approximately $\delta(1 - \cos \theta)$. A small deviation is cheap ($\mathcal{O}(\theta^2)$ cost) but displaces B laterally, breaking the collinearity that the cut-off requires.

B 's optimal sidestep is perpendicular to $\overline{BX}$, away from A 's side. Perpendicular motion dedicates all effort to evasion: progress toward X is zero, so no forward motion is wasted at a suboptimal angle. Any backward component (along $-\hat{u}$) is strictly worse: it keeps B on the line BX , so P^* remains between $B(t)$ and X and the cut-off still lands. Any forward+lateral mix loses lateral displacement compared to pure perpendicular. The unique optimum is 90° , on the side opposite A .

However, sidestepping has a limit. If A 's total time through the cut-off point and then onward to X is still less than B 's direct time, sidestepping cannot save B : even if the cut-off fails, A reaches X first.

Four Regimes

A 's time through P^* then to X is:

$$T_A^{\text{via}} = \tau_A + (1 - \lambda^*)d_B/v_A \quad (7)$$

Since the straight line is the shortest path, B 's time to reach X on any trajectory is at least T_B . If $T_A^{\text{via}} \leq T_B$, A wins regardless of B 's strategy.

The *guaranteed inside water condition* (Appendix A.3 C) simplifies to:

$$h\sqrt{\frac{1-r}{1+r}} \leq p - (1-r)d_B \quad (8)$$

Note: this condition alone is not sufficient — it presupposes that P^* exists, i.e., that the inside water condition 5 also holds. Regime ② requires both.

Theorem 2 (Four regimes). *Every configuration falls into exactly one regime:*

1. **A wins direct.** 3. A swims to X .
2. **Guaranteed inside water.** 3 fails, 5 holds, 8 holds. A swims to P^* : if B goes direct, A cuts off; if B sidesteps, A continues to X and still arrives first. B cannot change the outcome.
3. **Contested inside water.** 3 fails, 5 holds, 8 fails. A can cut off B -direct, but B can sidestep and still win the race. The game has a mixed Nash equilibrium: B randomises between direct and sidestep, A randomises between cutting off and direct.
4. **B wins.** 3 fails and 5 fails. B swims direct.

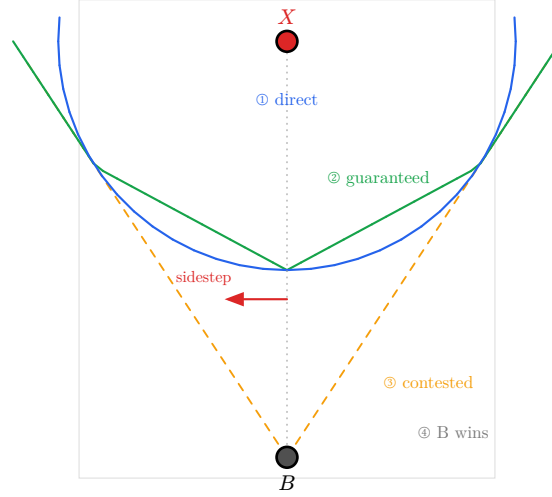


Figure 2: Four regimes for $r = 0.55$. ① Blue circle: A wins direct. ② Green boundary: guaranteed inside water — intersection of guaranteed and inside water conditions, where A wins regardless of B 's sidestep. ③ Between green and orange: contested — cutting off is a threat but B can sidestep. ④ Outside: B wins. Red arrow: B 's sidestep direction.

The Equal Speed Case ($r = 1$)

When $r = 1$, the interception angle equation $\cos \psi = 1$ forces $\psi = 0$: the inside water point collapses to X . The inside water wedge opens to 90° , but the direct-win circle covers the same region. The guaranteed condition reduces to $0 < p$, which is equivalent to the inside water condition itself.

Corollary. *When $r = 1$, the four regimes collapse to two: A wins (closer to X) or B wins (closer to X). There is no inside water and no sidestep.*

This is the default when speeds are unknown. Inside water exists only when there is a speed differential, and benefits only the slower side.

Strategy Collapse

Three candidate strategies: Direct (swim to X), Inside water (swim to P^*), Deep (iterated best response, taken to the limit). A priori $3^4 = 81$ combinations of strategy and assumption about the opponent. The collapse is partial:

In regimes ①, ②, ④, all strategies converge to a single pure decision. By Theorem 1, B has no cut-off option, so B 's direct is uniformly optimal. A 's pure best response is determined by the regime (direct in ①, inside water in ②, accept loss in ④). Iterated best response terminates in one step; reactive (continuous observation) yields the same straight-line trajectory.

In regime ③, the equilibrium is mixed. The iterated best response oscillates with period 2: from (direct, direct) $\rightarrow$ (inside water, direct) $\rightarrow$ (inside water, sidestep) $\rightarrow$ (direct, sidestep) $\rightarrow$ (direct, direct) $\rightarrow \dots$ The pure-strategy iteration does not converge. The static one-shot game has a unique mixed Nash equilibrium in which B randomises between direct and sidestep, and A randomises between inside water and direct. The reactive (continuous) strategy converges to this mixed equilibrium: the long-run frequencies of pure actions match the equilibrium probabilities.

Theorem 3 (Strategy collapse). *The 81 combinations reduce to the Nash equilibrium of the one-shot game. This equilibrium is pure in regimes ①, ②, ④, and mixed in regime ③. Iterated, reactive, and limit strategies converge to the same equilibrium; in regime ③, the play-by-play frequencies of pure actions converge to the mixed equilibrium probabilities.*

Interactive Companion

An interactive game is available at insidewater.xyz where the player can drag positions, choose speeds, and visualise the regime in real time. The implementation formulas are in Appendix C D.

Conclusion

Two players race to a goal. The slower can sometimes win by inside water; the faster never can. The faster player's recourse is the sidestep, but in the *guaranteed inside water* region even this fails. Four regimes emerge: direct win, guaranteed inside water, contested inside water, and loss. Complex multi-step strategies collapse to the Nash equilibrium of the one-shot game — pure in three regimes, mixed in the contested fourth.

Versus the classical Apollonius circle, our contributions are: the inside water asymmetry (Theorem 1), the four-regime classification (Theorem 2), and the strategy collapse (Theorem 3).

A Proof: the faster player cannot cut off

X at origin, $A = d_A(\cos \varphi, \sin \varphi)$, $B = (d_B, 0)$. A 's path: $Q(\mu) = (1 - \mu)A$, $\mu \in [0, 1]$. Arrivals: $\tau_A(\mu) = \mu d_A / v_A$, $\tau_B(\mu) = \|Q(\mu) - B\| / v_B$.

Step 1. Define $g(\mu) = \tau_B(\mu) - \tau_A(\mu)$. Since $\dot{Q} = -A$:

$$g'(\mu) = \frac{d_A}{v_B} \left[\cos \psi(\mu) - \frac{1}{r} \right] \quad (9)$$

where $\cos \psi(\mu) = (B - Q) \cdot \hat{A} / \|B - Q\| \in [-1, 1]$ and $\hat{A} = A / d_A$.

Step 2. Since $1/r > 1 \geq \cos \psi$, the bracket is always negative: $g'(\mu) < 0$ for all μ .

Step 3. g is strictly decreasing. Minimum at $\mu = 1$: $g(1) = T_B - T_A$.

Step 4. When $T_A < T_B$ (B wins the race), $g(1) > 0$, so $g(\mu) > 0$ for all $\mu \in (0, 1]$. B arrives later than A everywhere. $\square$

B Derivation of the inside water point

B 's direct path: $P(\lambda) = (1 - \lambda)B$, $\lambda \in [0, 1]$. Define $f(\lambda) = \|P(\lambda) - A\| / v_A - \lambda d_B / v_B$.

With $\xi = p - \lambda d_B$ (offset along $\overline{BX}$) and h (perpendicular distance):

$$f'(\lambda) = 0 \quad \Rightarrow \quad \frac{-\xi}{\sqrt{\xi^2 + h^2}} = r \quad (10)$$

Solving: $\xi^* = -rh / \sqrt{1 - r^2}$, giving:

$$\lambda^* = (p + sh) / d_B, \quad s = r / \sqrt{1 - r^2} \quad (11)$$

Arrival time: $\tau_A = h / (v_A \sqrt{1 - r^2})$.

The condition $\tau_A < \tau_B(\lambda^*)$ simplifies. Starting from $h / (v_A \sqrt{1 - r^2}) < (p + sh) / v_B$ and multiplying by $v_B = v_A / r$:

$$h / (r \sqrt{1 - r^2}) < p + rh / \sqrt{1 - r^2} \quad (12)$$

$$h(1 - r^2) / (r \sqrt{1 - r^2}) < p \quad (13)$$

$$h\sqrt{1-r^2}/r < p \Rightarrow h\sqrt{1-r^2} < rp \quad \square \quad (14)$$

C Guaranteed inside water derivation

A 's total time via P^* then to X : $T_A^{\text{via}} = h/(v_A\sqrt{1-r^2}) + (1-\lambda^*)d_B/v_A$.

Require $T_A^{\text{via}} \leq T_B = rd_B/v_A$. Multiply by v_A :

$$h/\sqrt{1-r^2} + d_B - p - rh/\sqrt{1-r^2} \leq rd_B \quad (15)$$

$$h(1-r)/\sqrt{1-r^2} \leq p - (1-r)d_B \quad (16)$$

$$h\sqrt{(1-r)/(1+r)} \leq p - (1-r)d_B \quad \square \quad (17)$$

Why sidestepping fails in regime ②: B 's time to X along any trajectory is at least T_B (the straight line is the shortest path). Since $T_A^{\text{via}} \leq T_B$, A reaches X before B regardless of B 's path.

Note on the conditions: The guaranteed condition does not imply the inside water condition. There are configurations where the algebraic guaranteed condition holds but no P^* exists in $(0, 1]$. In those configurations, A has no cut-off threat and the “guaranteed” condition is vacuous. Regime ② strictly requires both conditions to hold, plus the failure of 3.

D General coordinate formulas

For arbitrary A, B, X with $v_A \leq v_B$, $r = v_A/v_B$:

Projection:

$$\hat{u} = (X - B)/\|X - B\|, \quad d_B = \|X - B\|, \quad d_A = \|X - A\|$$

$$p = (A - B) \cdot \hat{u}, \quad \mathbf{n} = (A - B) - p\hat{u}, \quad h = \|\mathbf{n}\|$$

Direct:

$$A \text{ wins direct} \Leftrightarrow d_A < rd_B$$

Inside water point ($r < 1$):

$$s = r/\sqrt{1-r^2}, \quad \lambda^* = (p + sh)/d_B, \quad P^* = B + \lambda^* d_B \hat{u}$$

$$\tau_A = h/(v_A \sqrt{1-r^2})$$

Inside water condition:

$$h\sqrt{1-r^2} < rp \quad \text{and} \quad 0 < \lambda^* \leq 1$$

Guaranteed inside water condition:

$$h\sqrt{(1-r)/(1+r)} < p - (1-r)d_B$$

Swim direction vectors (assuming $h > 0$; the case $h = 0$ is collinear and degenerate):

Define the unit perpendicular to $\hat{u}$, pointing from line $\overline{BX}$ toward A :

$$\hat{n} = \mathbf{n}/h$$

With $A = B + p\hat{u} + h\hat{n}$ and $P^* = B + (p + sh)\hat{u}$:

$$P^* - A = sh\hat{u} - h\hat{n}, \quad \|P^* - A\| = h/\sqrt{1-r^2}$$

A 's swim direction (when going to P^*):

$$\hat{d}_A = r\hat{u} - \sqrt{1-r^2}\hat{n}$$

The forward component r and lateral component $-\sqrt{1-r^2}$ correspond to the Apollonius interception angle $\arccos(r)$ from $\hat{u}$, with the lateral tilt always toward line $\overline{BX}$ (i.e., in the $-\hat{n}$ direction, away from A 's current side). This angle is the same in every configuration where inside water is feasible; only $\hat{u}$ and $\hat{n}$ rotate with the geometry.

B 's sidestep direction:

$$\hat{d}_{\text{step}} = -\hat{n}$$

Pure perpendicular to $\hat{u}$, on the side opposite A .

Both lateral motions are in the same direction $-\hat{n}$. A and B both swim *away from A 's current side*, with A also moving forward toward the goal at scale r , and B moving purely sideways.

Outcome:

$$\text{result} = \begin{cases} \text{A wins (direct)} & \text{if } d_A < r d_B \\ \text{A wins (guaranteed)} & \text{if inside water and guaranteed both hold} \\ \text{contested (mixed NE)} & \text{if inside water holds but guaranteed fails} \\ \text{B wins} & \text{otherwise} \end{cases}$$

Cartesian view. For $X = (X_x, X_y)$, $B = (B_x, B_y)$, $A = (A_x, A_y)$, $r = v_A/v_B \in (0, 1]$:

$$d_B = \sqrt{(X_x - B_x)^2 + (X_y - B_y)^2}, \quad d_A = \sqrt{(X_x - A_x)^2 + (X_y - A_y)^2}$$

$$(u_x, u_y) = ((X_x - B_x)/d_B, (X_y - B_y)/d_B)$$

$$p = (A_x - B_x)u_x + (A_y - B_y)u_y$$

$$(n_x, n_y) = (A_x - B_x - pu_x, A_y - B_y - pu_y), \quad h = \sqrt{n_x^2 + n_y^2}$$

When $h > 0$, define $(\hat{n}_x, \hat{n}_y) = (n_x/h, n_y/h)$ (the case $h = 0$ is collinear; cut-off degenerates to the direct race).

Conditions:

$$\textcircled{1} \text{ direct: } d_A < r d_B$$

$$\textcircled{2} \textcircled{3} \text{ inside water: } h\sqrt{1-r^2} < rp \quad \text{and} \quad 0 < (p + sh)/d_B \leq 1, \quad s = r/\sqrt{1-r^2}$$

$$\textcircled{2} \text{ guaranteed: } h\sqrt{(1-r)/(1+r)} < p - (1-r)d_B$$

Cut-off point:

$$P^* = (B_x + (p + sh)u_x, B_y + (p + sh)u_y)$$

A's swim direction (when going to P^*):

$$\hat{d}_A = (ru_x - \sqrt{1-r^2}\hat{n}_x, ru_y - \sqrt{1-r^2}\hat{n}_y)$$

B's sidestep direction:

$$\hat{d}_{\text{step}} = (-\hat{n}_x, -\hat{n}_y)$$

To render on screen with the goal at $(1/2, 1)$ and the defender at $(1/2, 0)$, apply the unique similarity sending $X \mapsto (1/2, 1)$ and $B \mapsto (1/2, 0)$. All conditions, the cut-off point, and the swim directions are preserved under this transformation.